

 poolside/**Laguna-XS-2.1-FP8** 

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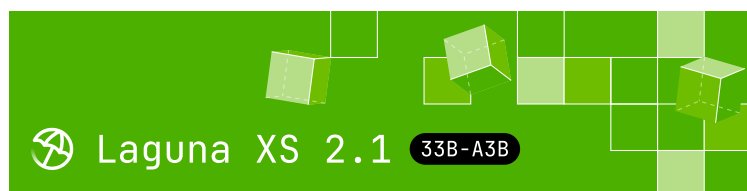
→ Copy to bucket **NEW**

 **Model card**

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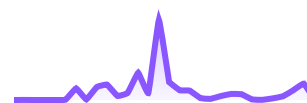
## Laguna XS 2.1-FP8

Laguna XS 2.1-FP8 is a 33B total parameter Mixture-of-Experts model with 3B activated parameters per token designed for agentic coding and long-horizon work on a local machine. It uses Sliding Window Attention with per-head gating in 30 out of 40 layers for fast inference and low KV cache requirements.

This is the FP8 variant with an FP8-quantized KV cache. The [BF16](#), [NVFP4](#) and [INT4](#) variants are also available on Hugging Face.

## Highlights

Downloads last month  
**9,544**



 **Safetensors** 

Model size 33B params

Tensor type F32 · BF16 · F8\_E4M3

 Chat template

 Files info

 **Inference Providers** **NEW**

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This model isn't deployed by any Inference Provider.

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 **Model tree for poolside/Laguna-X...**

Base model ..... poolside/Laguna-XS-2.1

 **Quantized** (22) ..... [this model](#)

Finetunes ..... [1 model](#)

 **Collection including poolside/Lag...**

**Laguna XS 2.1**  Collection

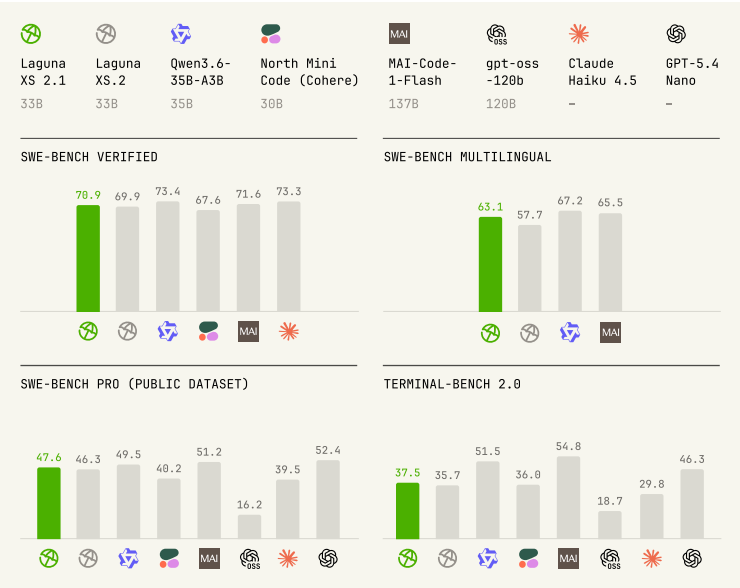
- **Mixed SWA and global attention layout:**  
Laguna XS 2.1 uses sigmoid gating with per-layer rotary scales, enabling mixed SWA (Sliding Window Attention) and global attention layers in a 3:1 ratio (across 40 total layers)
  - **KV cache in FP8:** KV cache quantized to FP8, reducing memory per token
  - **Native reasoning support:** Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
  - **Local-ready:** At 33B total parameters and 3B activated, Laguna XS 2.1 is compact enough to run on a Mac with 36 GB of RAM. Available on [Ollama](#) and [llama.cpp](#) (BF16 and Q4\_K\_M only)
  - **OpenMDW-1.1 license:** Use and modify the model and associated materials freely for commercial and non-commercial purposes ([learn more about OpenMDW](#))
- 

## Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 33B total with 3B activated per token
- Optimizer: Muon
- Layers: 40 layers (10 layers with global attention, 30 layers with sliding window attention)
- Experts: 256 experts with 1 shared expert

- Sliding Window: 512 tokens
- Modality: text-to-text
- Context window: 262,144 tokens
- Reasoning support: interleaved thinking with preserved thinking

Benchmark results



| Model           | Size (total params.) | SWE-bench Verified | SWE-bench Multilingual | SWE-Bench Pro (Public Dataset) |
|-----------------|----------------------|--------------------|------------------------|--------------------------------|
| Laguna XS 2.1   | 33B                  | 70.9%              | 63.1%                  | 47.6%                          |
| Laguna XS.2     | 33B                  | 69.9%              | 57.7%                  | 46.3%                          |
| Qwen3.6-35B-A3B | 35B                  | 73.4%              | 67.2%                  | 49.5%                          |

| Model                    | Size<br>(total<br>params.) | SWE-<br>bench<br>Verified | SWE-bench<br>Multilingual | SWE-<br>Bench<br>Pro<br>(Public<br>Dataset) |
|--------------------------|----------------------------|---------------------------|---------------------------|---------------------------------------------|
| North<br>Mini<br>Code    | 30B                        | 67.6%                     | -                         | 40.2%                                       |
| MAI-<br>Code-1-<br>Flash | 137B                       | 71.6%                     | 65.5%                     | 51.2%                                       |
| gpt-oss-<br>120B         | 120B                       | -                         | -                         | 16.2%                                       |
| Claude<br>Haiku 4.5      | -                          | 73.3%                     | -                         | 39.5%                                       |
| GPT-5.4<br>Nano          | -                          | -                         | -                         | 52.4%                                       |

*We used the highest publicly-referenced scores for all comparison models across each benchmark. In all cases these were official scores published in release blog posts or equivalent, with the exception of gpt-oss-120b and Claude Haiku 4.5 where the highest published (verified) scores for SWE-Bench Pro and Terminal-Bench 2.0 are from their respective official leaderboards.*

► Expand for benchmarking methodology

Quantization quality

Laguna XS 2.1-FP8 was evaluated against the unquantized BF16 checkpoint on the same four agentic benchmarks. Scores are mean pass@1; the ±

figure is the run-to-run variation and *RH* is the share of runs flagged by our post-hoc reward-hack judge (see the benchmarking methodology note above).

| Benchmark                      | Laguna XS 2.1 (BF16)         | Laguna XS 2.1-FP8            |
|--------------------------------|------------------------------|------------------------------|
| SWE-bench Verified             | 70.85% ± 0.85%<br>(6.51% RH) | 70.75% ± 0.85%<br>(6.22% RH) |
| SWE-bench Multilingual         | 63.17% ± 1.42%<br>(7.25% RH) | 64.92% ± 1.42%<br>(6.80% RH) |
| SWE-Bench Pro (Public Dataset) | 47.61% ± 0.96%<br>(2.40% RH) | 48.02% ± 0.89%<br>(2.67% RH) |
| Terminal-Bench 2.0             | 37.53% ± 2.81%<br>(2.12% RH) | 40.22% ± 2.70%<br>(3.02% RH) |

## Usage

Laguna XS 2.1-FP8 has launch-day support in vLLM, SGLang and Transformers, and TRT-LLM thanks to the support of the team at NVIDIA.

For complete usage instructions, see the main [Laguna XS 2.1 model card](#).

## Local deployment

Laguna XS 2.1-FP8 is supported in vLLM, SGLang and Transformers, and TRT-LLM thanks to the support of the team at NVIDIA. Use Laguna XS 2.1 with Ollama (with MLX support) or Llama.cpp (BF16 and Q4\_K\_M only) for the best results on your local machine.

## vLLM

The full vLLM recipe is on the main [Laguna XS 2.1 model card](#) and on the [vLLM recipes page](#).

Quantization is detected automatically from `quantization_config` in this checkpoint, so the same command works with `poolside/Laguna-XS-2.1-FP8` substituted for the model ID. Set `VLLM_BLOCKSCALE_FP8_GEMM_FLASHINFER=0` when serving with vLLM.

Tool calling requires a vLLM build with [vllm-project/vllm#47311](#); on older builds pass `--tool-call-parser glm47` (the GLM 4.7 parser) instead of `poolside_v1`.

The FP8-quantized KV cache requires vLLM `>= 0.22.0`. Earlier versions produce scrambled output on non-Hopper GPUs because of a per-layer attention-head count bug, fixed in [vllm#42650](#). On older vLLM, disable the FP8 KV cache by adding `--kv-cache-dtype-skip-layers $(seq 0 39)`.

## SGLang

Laguna XS 2.1 is supported in SGLang via [sgl-project/sglang#24204](#). Quantization is detected automatically from `quantization_config`, so no extra flags are required. See the [SGLang cookbook entry](#) and the main [Laguna XS 2.1 model card](#) for a serving recipe.

## Transformers

The full Transformers recipe is on the main [Laguna XS 2.1 model card](#). Substitute `poolside/Laguna-XS-`

2.1-FP8 for the model ID; quantization is detected automatically from `quantization_config`.

## TRT-LLM

Laguna XS 2.1 support ships in TensorRT-LLM `>=1.3.0rc16`; see the install recipe on the main [Laguna XS 2.1 model card](#). Substitute `poolside/Laguna-XS-2.1-FP8` for the model ID; quantization is detected automatically from `quantization_config`, no extra flags required.

```
from tensorrt_llm import LLM

llm = LLM(model="poolside/Laguna-XS-2.1-FP8")
```

## Ollama

Available on the [Ollama library](#).

**macOS (Metal) users:** Chat (`ollama run /api/chat`) works as expected on Linux/CUDA. On macOS/Metal it may currently return empty output; the root cause is not yet fully understood and we're investigating it with the Ollama team. On a Mac, use a Linux/CUDA host, or the `/api/generate` endpoint with `"raw": true`.

## Controlling reasoning

Laguna XS 2.1-FP8 uses the same reasoning controls (interleaved thinking, preserved reasoning, and the `enable_thinking` flag) as the base model. See the [Controlling reasoning](#) section of the main Laguna XS 2.1 model card.

## License

This model is licensed under the [OpenMDW-1.1 License](#).

### Intended and Responsible Use

Laguna XS 2.1-FP8 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna XS 2.1-FP8 is subject to the [OpenMDW-1.1 License](#), and should be

 System theme with [Poolside's Acceptable Use Policy](#). We advise against circumventing Laguna XS 2.1-FP8 safety guardrails without implementing substantially equivalent mitigations appropriate for your use case.

Please report security vulnerabilities or safety concerns to [security@poolside.ai](mailto:security@poolside.ai).



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